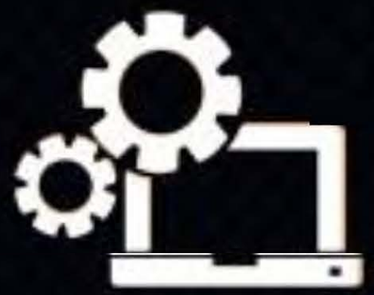
CS & IT ENGINEERING

DIGITAL LOGIC

Minimization



Lecture No. 2



By - CHANDAN SIR

Recap of Previous Lecture



Boolean Algebra

Basic Terms

Topics to be Covered



Boolean Algebra

Theorems

Question



Topic : Boolean Algebra

① Distribution Theorem :→

$$(A+B)(A+C)$$

$$\Rightarrow A\cdot A + AC + AB + BC$$

$$\Rightarrow A + AC + AB + BC$$

$$\Rightarrow A[1+C+B] + BC$$

$$\Rightarrow A + BC$$

$$A+BC = (A+B)\cdot(A+C)$$

$$A+BCD = (A+B)(A+C)\cdot(A+D)$$

Q $A+\bar{A}B$

$$(A+\bar{A})(A+B) = (A+B)$$

② Redundancy Theorem :→ {Concensus Theorem}

$AB + \bar{A}C + BC$

$AB + \bar{A}C + (\bar{A}+A)BC$

$AB + \bar{A}C + \bar{A}BC + ABC$

$AB[1+C] + \bar{A}C[1+B]$

$AB + \bar{A}C$

Q $\bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{C}$

$\bar{A}\bar{B} + B\bar{C}$

Redundant = $\bar{A}\bar{C}$

Q $\bar{A}BC + ACD + BCD$

$BC \cdot CD$

BCD

$= \bar{A}BC + ACD$

$AB + A\bar{C} + BC$

$A\bar{C} + BC$

Ans

$AB + \bar{A}C + B\bar{C}$

$AB + \bar{A}C$

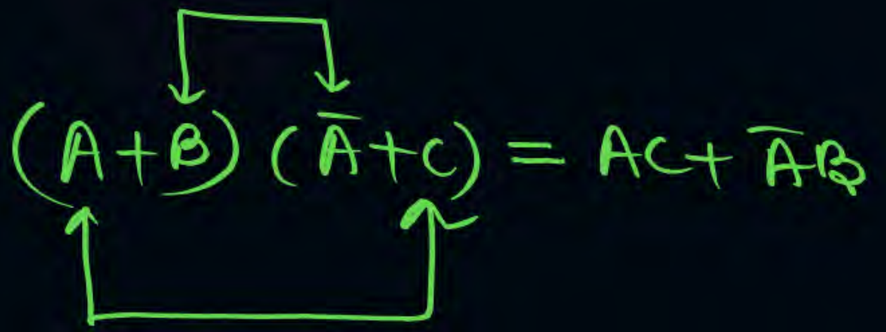
③ Transpose Theorem :→

$$(A+B)\cdot(\bar{A}+C)$$

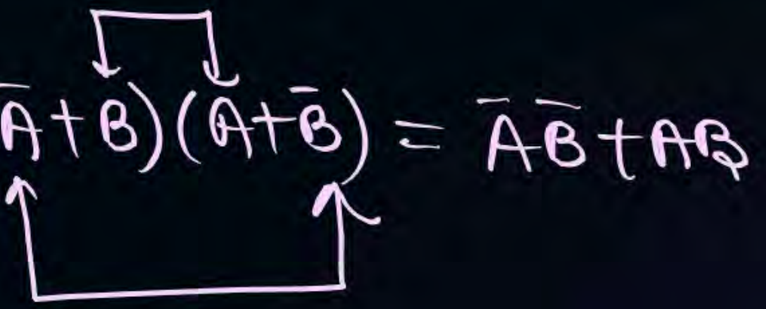
$$\Rightarrow A\cdot\bar{A}+AC+\bar{A}B+BC$$

$$\Rightarrow AC+\bar{A}B+BC$$

$$\Rightarrow AC+\bar{A}B$$



Ex ≡



④ D- Morgan Law

$$\overline{ABC} = \bar{A} + \bar{B} + \bar{C}$$

$$\overline{A+B+C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

$$A+BC=(A+B)(A+C)$$

$$AB+\bar{A}C+BC=AB+\bar{A}C$$

$$(A+B)(\bar{A}+C)=AC+\bar{A}B$$

$$\overline{ABC}=\bar{A}+\bar{B}+\bar{C}$$

$$\overline{A+B+C}=\bar{A}\cdot\bar{B}\cdot\bar{C}$$

#Q. Find the minimum number of the NAND gate required to implement the Boolean function given below:

$f(A, B, C) = A + ABC + AB\bar{C}$

$= A[1 + BC + B\bar{C}]$

$= A$





#Q. Minimize the expression:

$f(A, B) = A + A\bar{B}$

$= A[1 + \bar{B}]$

$= A$

$A + A\bar{B}$

$A[1 + \bar{B}]$

$A \cdot 1 = A$

#Q. Minimize the expression:

$f(A, B) = \overline{A}\,\overline{B} + \overline{A}B + AB$

$= \overline{A}[\overline{B} + B] + AB$

$= \overline{A} + AB$

$= (\overline{A} + A)(\overline{A} + B)$

$= \overline{A} + B$

#Q. Minimize the expression.

$f(A, B) = \overline{A}\,\overline{B} + \overline{A}B + A\overline{B} + AB$

$= \overline{A}[\overline{B} + B] + A[\overline{B} + B]$

$= \overline{A} + A$

$= 1$

Note 1. If in the function all minterms are present then minimal form will be 1.

2. If in the function all maxterms present then minimal form will be zero.

$n=1$ (A)

A
$\bar{A}$
0
1

} 4

$n=2$ (A, B)

$\bar{A}\bar{B}$	$\bar{A}+\bar{B}$
$\bar{A}B$	$\bar{A}+B$
$A\bar{B}$	$A+\bar{B}$
AB	$A+B$

A	0
$\bar{A}$	1
B	$\bar{A}B+A\bar{B}$
$\bar{B}$	$\bar{A}\bar{B}+AB$

16

minterm

4

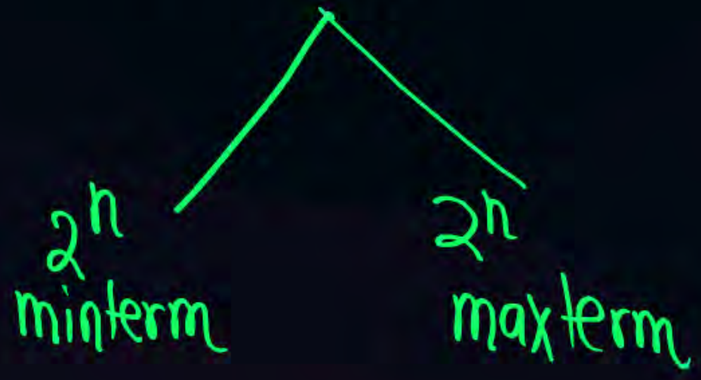
maxterm

4

'n' Variable

Total no of minimized Expression

$$= 2^{2^n}$$



#Q. Minimize the expression.

$f(A, B) = \bar{A}B + A\bar{B}$

$= A \oplus B$

already minimized

$\bar{A}\bar{B} + AB = A \odot B$

001 010 100 111

$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC = A \oplus B \oplus C$

$\Sigma m(1, 2, 4, 7)$

#Q. Minimize the expression.

$f(A, B) = AB + \bar{A}C + BC$

(A) $AB + \bar{A}C$

(B) $AB + BC$

(C) BC

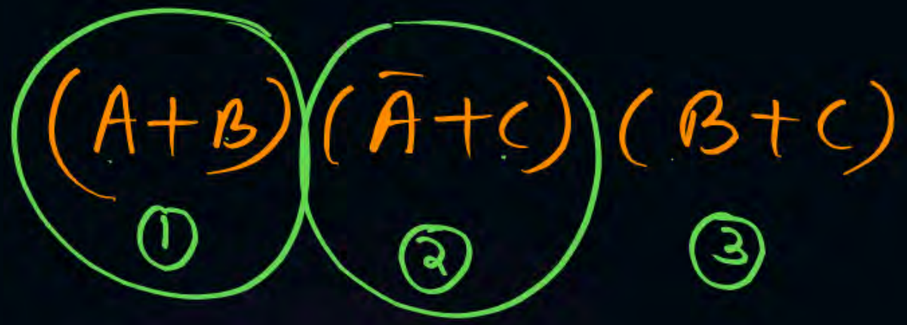
(D) Wrong format

$f(\ )$ ← Variables

#Q. Minimize the expression.

$f(A, B, C) = \overline{A}\,\overline{B} + \overline{A}C + \overline{B}\,\overline{C}$

$= \overline{A}C + \overline{B}\,\overline{C}$



$(A+B)(\bar{A}+C)$

$AC+\bar{A}B$

#Q. Minimize the expression.

$f(A, B, C) = (A + B)(A + C)(\overline{B} + C)$

$= (A+B)(\overline{B}+C)$

$= (B+\overline{A})(\overline{B}+C)$

$= BC + A\overline{B}$

#Q. Write the function for truth table and minimize it.

A	B	Y(O/p)
0	0	1
0	1	0
1	0	1
1	1	1

SOP

$\bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 0 + A\bar{B}\cdot 1 + AB\cdot 1$

$\bar{A}\bar{B} + A\bar{B} + AB$

$\bar{A}\bar{B} + A(\bar{B}+B) = A + \bar{A}\bar{B} = (A+\bar{A})(A+\bar{B}) = (A+\bar{B})$

POS

$f = (A+B+1)\cdot(A+\bar{B}+0)\cdot(\bar{A}+B+1)\cdot(\bar{A}+\bar{B}+1)$

$= 1\cdot(A+\bar{B})\cdot 1\cdot 1$

$= A+\bar{B}$ Ans

#Q. Write the function for truth table and minimize it.

A	B	Y(O/p)
0	0	C
0	1	$\overline{C}$
1	0	1
1	1	1

SOP

$y = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B} + AB$

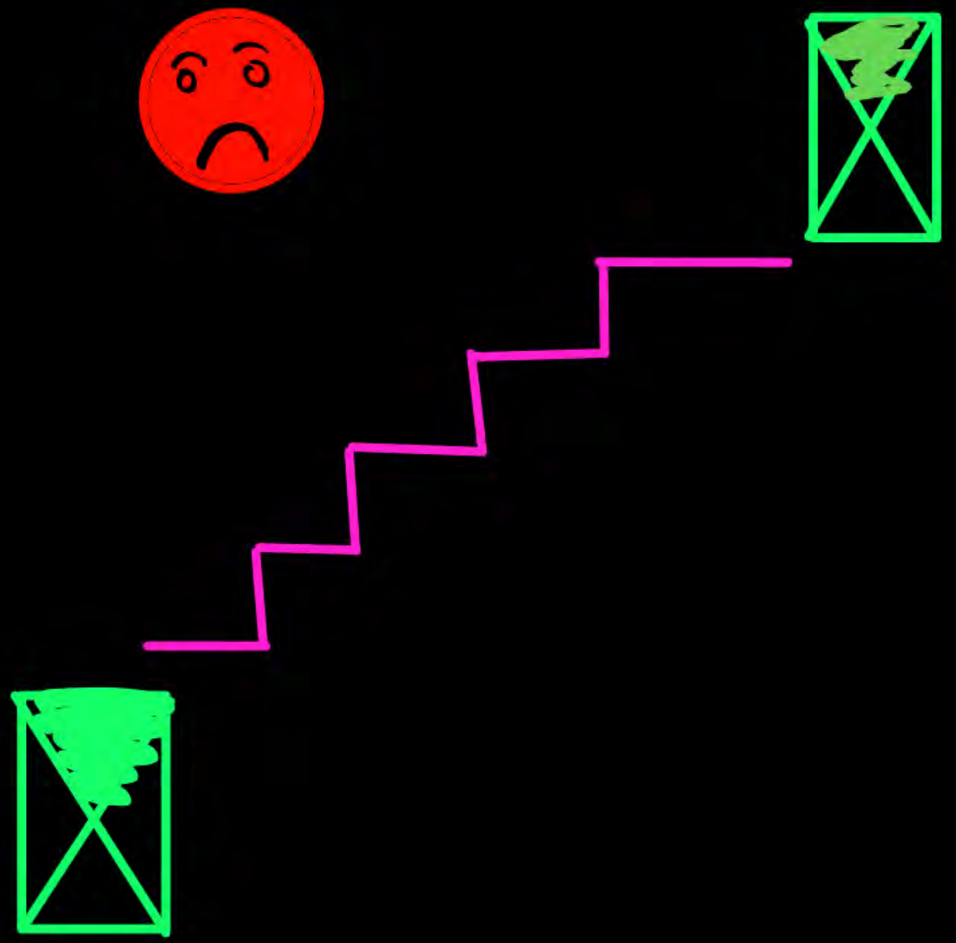
#Q. Two way switch is an example of which logic?

A AND

B OR

C X-OR

D NAND



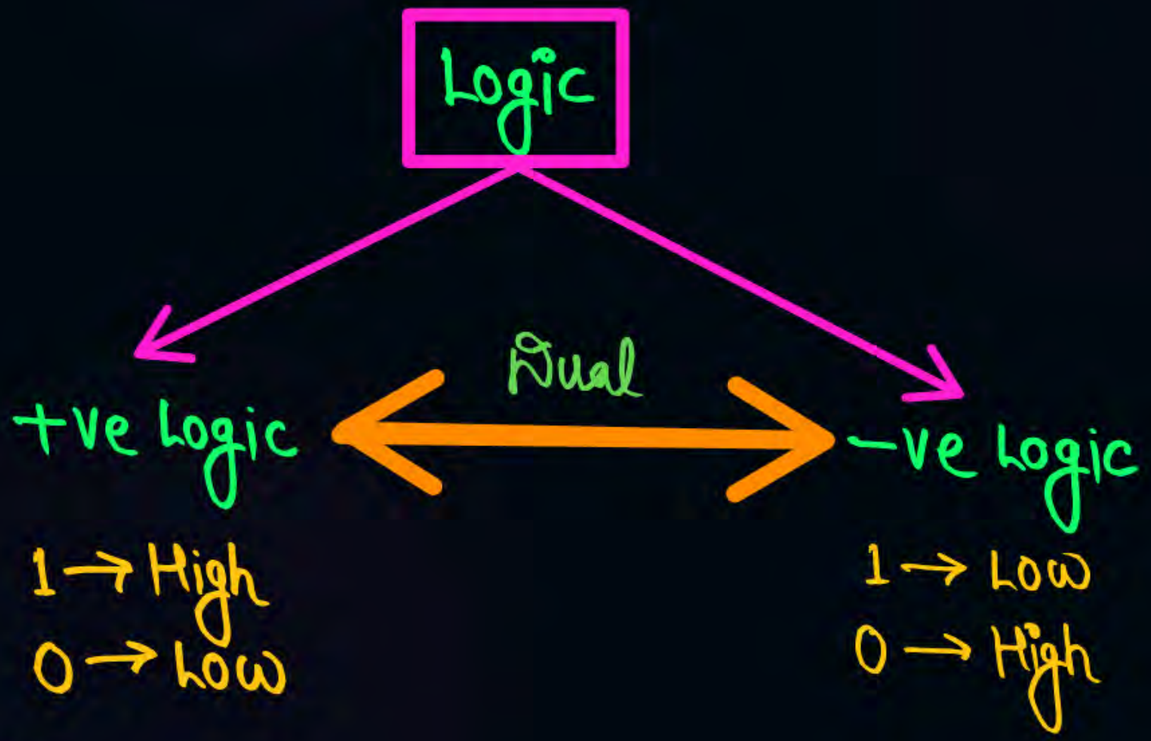
#Q. If we have '4' variable, then total different expression will be?

2^{2^n}

2^{2^4}

$2^{16} = 2^6 \cdot 2^{10}$

$= 64k$



Ex. 1 → 10V, 0 → 5V } +ve

Ex 1 → 0V, 0 → -5V } +ve

Ex 1 → -8V, 0 → -10V } +ve

Ex. 1 → 5V, 0 → 10V } -ve

Ex. 1 → 0V, 0 → +5V } -ve

Ex. 1 → -10V, 0 → -5V } -ve

NAND $\xleftrightarrow{\text{Dual}}$ NOR

$\overline{A \cdot B}$ $\xleftrightarrow{\text{Dual}}$ $\overline{A + B}$

X-OR $\xleftrightarrow{\text{Dual}}$ X-NOR

Buffer $\xleftrightarrow{\text{Dual}}$ Buffer

Inverter $\xleftrightarrow{\text{Dual}}$ Inverter

$A \xleftrightarrow{\text{Dual}} A$

$\overline{A} \xleftrightarrow{\text{Dual}} \overline{A}$

$A \cdot 1 \xrightarrow{\text{Dual}} ?$

$A \cdot 1 \longrightarrow A + 0$

$A \xrightarrow{\text{Dual}} A$

Q $f = AB + CDE$

$f^{\bar{N}} = (A+B) \cdot (C+D+E)$

Q $f = \overline{AB} + \overline{CDE}$

$f^{D} = (\overline{A+B}) \cdot (\overline{C+D+E})$

Q $f = A\bar{B}\bar{C} + D\bar{E}F + G\bar{H}$

$f^D = (A+\bar{B}+\bar{C})\cdot(D+\bar{E}+F)\cdot(G+\bar{H})$

$f = f^D$

Self Dual

$f = A$

$f^D = A$

Ex



Self Dual



2 Minute Summary

AND ⟷ (Dual) ⟷ OR

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

$0 \xrightarrow{\text{Dual}} 1$

$1 \xrightarrow{\text{Dual}} 0$

$A \cdot B \xrightarrow{\text{Dual}} A + B$

$A + B \xrightarrow{\text{Dual}} A \cdot B$

AND $\xrightarrow{\text{Dual}}$ OR

$\cdot \longleftrightarrow +$

Thank You GW Soldiers!

